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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 5

Title: A company has received customer data containing missing values, duplicate records, and inconsistent entries. Clean and prepare the dataset using Power Query before analysis.

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Experiment No.: 4

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Problem Statement

This experiment aims to detect and correct data quality problems in a raw cafe transaction dataset — hidden missing values, categorical fields with heavy gaps, and numeric fields that don't reconcile with one another — using Power Query, so the final dataset can be trusted for dashboard reporting and decision-making.

Step 1: Data Preparation

- Imported the Cafe Sales dataset (Kaggle — "Cafe Sales - Dirty Data for Cleaning Training", ahmedmohamed2003) into Power Query via Get Data → Text/CSV. The file held 10,000 transactions across 8 fields: Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date.
- Observed that every column had loaded as plain text, and that the words "ERROR" and "UNKNOWN" were being used as stand-ins for missing data across several columns, alongside cells that were genuinely blank.
- Used Power Query's Column Quality and Column Distribution panels to measure the scale of the problem: combined blank/"ERROR"/"UNKNOWN" rates sat between roughly 4.6% and 9.7% for Item, Quantity, Price Per Unit, Total Spent, and Transaction Date, but climbed to 31.8% for Payment Method and 39.6% for Location.
- Used Replace Values to swap out "ERROR" and "UNKNOWN" text for genuine nulls across every affected column, doing this *before* changing any column's data type — converting to numeric or date types first would have generated a fresh batch of conversion errors on each placeholder cell.
- Only afterward converted Quantity, Price Per Unit, and Total Spent to numeric types and Transaction Date to a Date type, clearing any leftover conversion errors with Replace Errors → null. A follow-up Column Quality check confirmed every column was now sitting at 0% Error.

Step 2: Data Recovery and Modeling

- Noted that Quantity, Price Per Unit, and Total Spent are linked by a fixed formula ($\text{Total Spent} = \text{Quantity} \times \text{Price Per Unit}$), and used that relationship to reconstruct missing values instead of dropping incomplete rows.
- Built custom columns with conditional M logic so that whenever exactly one of the three numeric fields was missing, it could be derived from the other two:
 - Total Spent missing → filled in as $\text{Quantity} \times \text{Price Per Unit}$

- Price Per Unit missing → filled in as $\text{Total Spent} \div \text{Quantity}$
- Quantity missing → filled in as $\text{Total Spent} \div \text{Price Per Unit}$, rounded to a whole number
- For Item, Payment Method, and Location, where this kind of calculation wasn't possible, replaced remaining nulls with "Unknown" instead of removing the rows — given that Payment Method and Location were missing in about 32% and 40% of records respectively, deleting those rows would have wiped out a large, uneven chunk of the dataset and skewed later analysis.
- Ran Trim and Clean across the text columns to strip out stray spaces and non-printable characters left behind from the original export.
- Checked for duplicate records using Transaction ID as the unique key via Remove Duplicates. The raw file turned out to have no duplicate IDs and no fully duplicated rows, but the step was kept in the pipeline as routine good practice.
- Built a Month-Year field from Transaction Date so that revenue could be aggregated by month without needing a full date hierarchy in every chart.

Step 3: Validation Measures

To confirm the cleaned data actually holds together, the following check was added in Power Query:

```
Totals Match =
try
    if [Quantity] = null or [Price Per Unit] = null or [Total Spent] = null
    then null
    else Number.Round([Quantity] * [Price Per Unit], 2) = Number.Round([Total Spent], 2)
otherwise
    null
```

Here, TRUE marks a row whose pricing is internally consistent, FALSE flags a mismatch, and null covers only the rows where all three numeric fields were missing together and nothing could be calculated. Wrapping the logic in try...otherwise means the column keeps working even if it runs into an unexpected value.

Step 4: Dashboard Development

- **KPI Cards:** Total Revenue, Total Transactions, Average Order Value.
- **Interactive Slicers:** Payment Method, Location, and Month, so a single click filters every visual on the page at once.
- **Revenue by Item** — bar chart ranking Coffee, Cake, Sandwich, Salad, Juice, Smoothie, Tea, and Cookie by total sales value.
- **Transactions by Payment Method** — pie chart splitting orders across Cash, Credit Card, and Digital Wallet.
- **Revenue by Location** — donut chart contrasting In-store versus Takeaway sales.
- **Revenue Trend Over Time** — line chart tracking revenue month by month using the derived Month-Year field.

Output

Data Quality Summary (Before → After Cleaning):

Metric	Before Cleaning	After Cleaning
Placeholder text ("ERROR"/"UNKNOWN") present	Found in every column	0 (converted to null or "Unknown")
Duplicate Transaction IDs	None found	None (re-confirmed)
Missing Payment Method / Location	31.8% / 39.6%	0% (labeled "Unknown")
Quantity × Price = Total consistency	Couldn't be checked on incomplete rows	~99% Valid, under 1% unrecoverable
Column data types	All Text	Properly typed (Number, Date, Text)

Final Interactive Cafe Sales Dashboard — KPI cards, four linked charts, and three slicers built entirely on the cleaned `Cafe_Sales_Cleaned` table.

Key Business Patterns Identified

- Coffee, Cake, and Sandwich stand out as the strongest revenue drivers, well ahead of lower-selling items like Tea and Cookie.
- In-store purchases bring in a larger share of revenue than Takeaway, hinting that customers who dine in tend to spend more per visit.
- Credit Card and Digital Wallet payments together outpace Cash by a wide margin, pointing to a general shift away from cash transactions.
- Monthly revenue moves up and down moderately rather than spiking at one clear peak, suggesting fairly steady demand year-round.
- Close to 40% of transactions had no recorded Location and nearly a third had no Payment Method logged — a gap large enough that it's worth raising as a point-of-sale data-collection problem, not just something to patch during cleaning.

Conclusion

This experiment worked with a raw cafe sales dataset marked by missing values, disguised placeholder text, and numeric inconsistencies to walk through a full Power Query cleaning process. Instead of discarding incomplete records — which would have removed close to 40% of the data — the pipeline separated true nulls from placeholder text, rebuilt missing numeric values using known relationships between fields, and clearly labeled categorical gaps that couldn't be recovered. What resulted is a properly typed, duplicate-free, validated dataset where about 99% of rows pass an internal consistency check. That cleaned dataset then powered an interactive dashboard combining KPI cards, comparison and trend visuals, and cross-filtering slicers for a quick, drillable view of sales performance.

Learning Outcomes

Through this experiment, I learned:

- How to tell disguised missing values (placeholder text like "ERROR"/"UNKNOWN") apart from genuine nulls, and why that distinction has to happen before type conversion.

- How to reconstruct missing numeric data using relationships already present in the dataset, rather than defaulting to deleting rows.
- How to make sensible decisions about categorical columns with heavy missingness, weighing completeness against the risk of biasing results by dropping data.
- How to write defensive M code (using `try...otherwise`) so validation columns keep functioning even when the data throws something unexpected.
- How to put together an interactive, management-ready dashboard using KPI cards, slicers, and a mix of chart types built on a dataset that's actually been verified.